CONSUMER FINANCE COMPANY

Loan Default EDA Case Study

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OVERVIEW

Outline

- Business problem and objectives
- Dataset overview and data cleaning
- Univariate analysis (who are our applicants?)
- Bivariate analysis (categorical variables, age, numeric variables, region rating)
- Previous application history (does past behaviour predict default?)
- Key findings and recommendations

CONTEXT

Business Problem

When a loan application is received, the company must decide whether to approve it.
But there are 2 types of risk:

Approving a bad loan

Applicant is likely to default → financial loss for the company

Rejecting a good loan

Applicant would have repaid → missed business opportunity

Goal: Use EDA to identify which applicant characteristics are associated with higher default risk, so the company can make more informed lending decisions.

DATA

Dataset Overview

FILE	ROWS	COLUMNS	DESCRIPTION
application_data.csv	307,511	122	Current loan application info
previous_application.csv	1,670,214	37	Past application history
columns_description.xlsx	–	–	Data dictionary

Target variable: 0 = no payment difficulty, 1 = payment difficulty (defaulted)

91.9%
No default (TARGET = 0)

8.1%
Default (TARGET = 1)

The dataset is highly imbalanced: about 11.4 non-defaulters for every 1 defaulter.

DATA

Data Cleaning

Missing values

Columns with more than 50% missing data were dropped before analysis.

DAYS_EMPLOYED anomaly

The value 365,243 (approximately 1,000 years) was a placeholder for unemployed applicants, so we replaced it with NaN.

New derived columns

Created `AGE`, `YEARS_WORKING`, and `CREDIT_INCOME_RATIO` from existing columns.

UNIVARIATE ANALYSIS

Who Are Our Applicants?

Contract & Gender

Most applicants are female and apply for cash loans.

Top Income Types

Important note on class imbalance

Only 8.1% of applicants defaulted. Because of this imbalance, looking at raw counts is misleading. Throughout this analysis we look at **default rates** (what % within each group defaulted) rather than counts. Income, credit, and annuity amounts are all right-skewed with a few very high values.

BIVARIATE ANALYSIS

Numeric Variables & Default Risk

We used box plots (capped at the 99th percentile to remove outliers) and group means to compare defaulters vs non-defaulters across four numeric variables.

VARIABLE	NON-DEFAULTERS (AVG)	DEFAULTERS (AVG)	TAKEAWAY
Annual Income	Higher	Lower	Defaulters earn less on average
Credit Amount	Slightly higher	Slightly lower	Weak difference; loan size alone not predictive
Annuity Amount	Similar	Similar	Little difference between groups
Credit / Income Ratio	Lower	Higher	Defaulters borrow more relative to their income

CREDIT_INCOME_RATIO is the most useful numeric signal. Applicants who borrow a larger multiple of their annual income are more likely to default. It captures what income and credit alone cannot.

BIVARIATE ANALYSIS

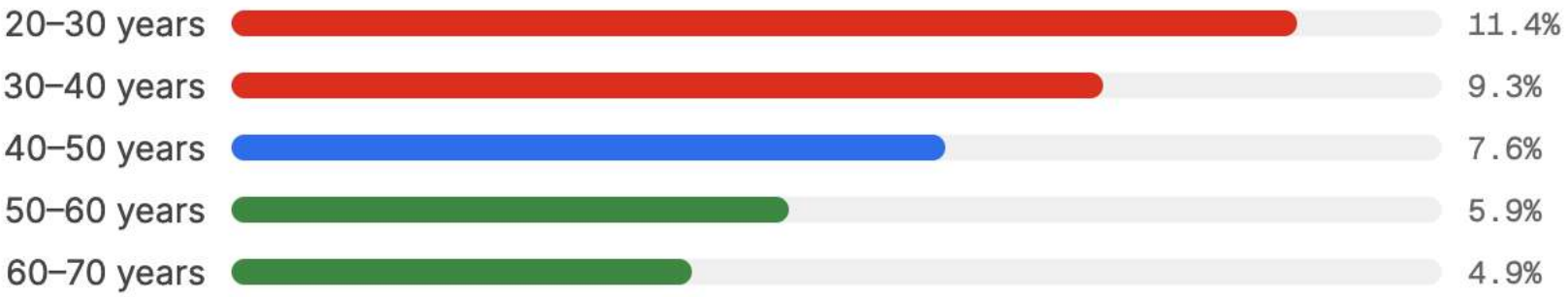
Key Categorical Variables

VARIABLE	HIGHER RISK GROUP	RATE	LOWER RISK GROUP	RATE
Education	Lower secondary	10.9%	Academic degree	1.8%
Income Type	Unemployed / Maternity leave	36-40%	Pensioner / State servant	5.4-5.8%
Housing Type	Rented apartment	12.3%	House / apartment	7.8%

BIVARIATE ANALYSIS

Age & Default Risk

We converted `DAYS_BIRTH` into age in years (flipping the sign, since the column is negative) and grouped applicants into 10-year bins. A reference line marks the overall average (8.1%).



Age has a very clear and consistent relationship with default risk. Younger applicants are significantly riskier: the 20-30 age group defaults at more than twice the rate of the 60-70 group.

BIVARIATE ANALYSIS

Region Rating

The dataset includes a region rating (1 = best, 3 = worst) for where the applicant lives. We looked at whether this predicts default risk.

Why does region matter?

Lower-rated regions tend to have weaker local economies and fewer job opportunities, making it harder for residents to keep up with loan payments.

Strong signal

Region 3 applicants default at more than twice the rate of Region 1 applicants (11.1% vs 4.8%), a meaningful gap.

PREVIOUS APPLICATION HISTORY

Does Past Behaviour Predict Default?

We merged the previous application dataset and computed each applicant's refusal rate (how often their past applications were refused).

Applicants with the highest past refusal rates default at over twice the rate of those with no prior refusals. If previous lenders said no, it was probably for good reason.

First-time vs returning applicants

Interestingly, first-time applicants actually default *less* than returning ones. Returning applicants who were previously refused pull the average up.

FINDINGS

Key Findings & Recommendations

High Risk Indicators

- Young age (20-30 years): 11.4% default rate
- Unemployed or on maternity leave: 36-40%
- Lower secondary education: 10.9%
- Renting or living with parents: 12.3%
- High past refusal rate: 15%+
- Lives in Region 3 (worst-rated): 11.1%
- High credit-to-income ratio

Lower Risk Indicators

- Older applicants (60s): 4.9% default rate
- Pensioners and state servants: 5-6%
- Academic degree holders: 1.8%
- No history of past refusals: 7.0%
- Lives in Region 1 (best-rated): 4.8%
- Low credit-to-income ratio

No single variable is sufficient on its own. The best approach is to look at a combination of employment status, age, education, housing, region, and past application history together.

Thank You